present this side of the story here: it is quantum gravity that I would like to get on to. The standard model is completed by the 1970s. There are approximately fifteen fields, whose quanta are the elementary particles (electrons, quarks, muons, neutrinos, Higgs, and little else), plus a few fields similar to the electromagnetic one, which describe electromagnetic force and the other forces operating at a nuclear scale, whose quanta are similar to the photons.

The standard model was not taken very seriously at first, due to its somewhat cobbled-together aspect, so different from the airy simplicity of general relativity and Maxwell's or Dirac's equations. Against expectations, however, all of its predictions have been confirmed. For more than thirty years, every single experiment of particle physics has done nothing but repeatedly reconfirm the standard model. A recent confirmation was the discovery of the Higgs particle, which caused a sensation in 2013. Introduced to render the theory coherent, the Higgs field seemed a bit artificial – until the Higgs particle, the quantum of this field, was actually observed and found to have precisely the properties predicted by the standard model.[fn25] (The fact that it has been called 'the God particle' is so stupid as to be unworthy of comment.) In short, despite its unjustly modest name, the standard model has been a triumph.

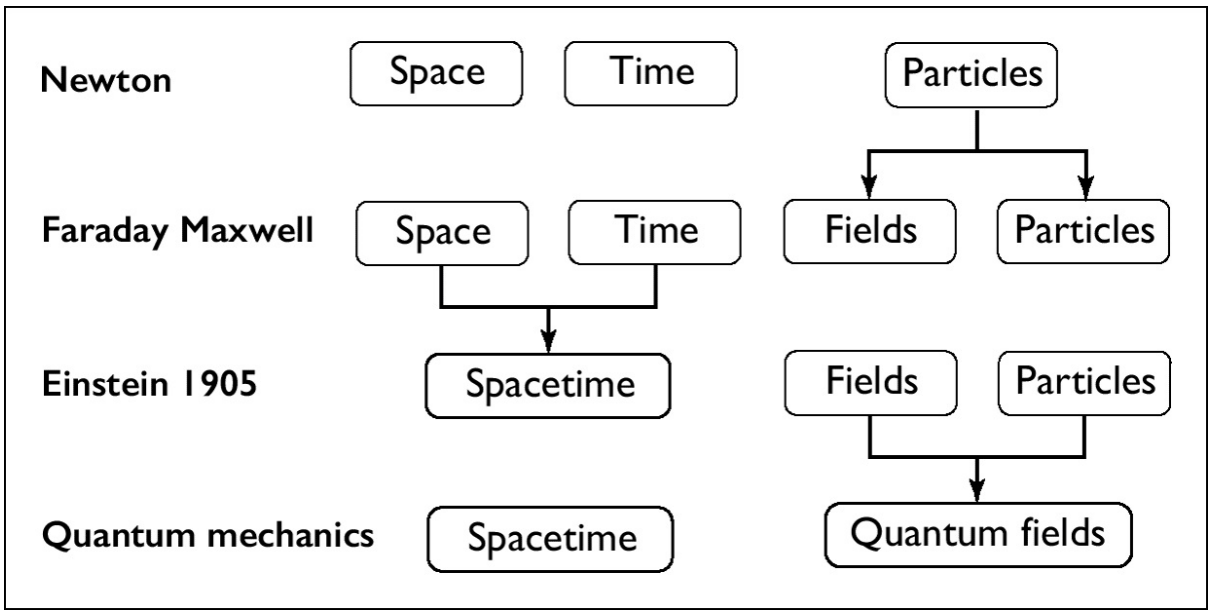


Figure 4.6 What is the world made of?

Quantum mechanics, with its fields/particles, offers today a spectacularly effective description of nature. The world is not made up of fields and particles but of a single type of entity: the quantum field. There are no longer particles which move in space with the passage of time, but quantum fields